LAB ASSIGNMENT
& SUBMISSION WORKSHEET

DATABASE MANAGEMENT SYSTEMS (DBMS) LAB

Lab Session	Lab 1 – Installation, Basic Setup & DML Operations (INSERT, UPDATE, DELETE)
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Submission note: This document contains the SQL statements and expected outputs for all five tasks. Run the statements in MySQL and compare your actual output with the expected results before submission.

Prepared for academic lab submission

Instructions & Database Setup

Open MySQL Command Line Client and execute the statements in order. If the database already exists, use the alternative CREATE DATABASE IF NOT EXISTS statement shown below.

```
CREATE DATABASE IF NOT EXISTS student_lab_db;
USE student_lab_db;
```

Task 1: Table Creation & Setup (DDL)

Q1. Create the Products table and describe its structure.

```
CREATE TABLE Products (
    ProductID INT PRIMARY KEY AUTO_INCREMENT,
    ProductName VARCHAR(50) NOT NULL,
    Category VARCHAR(30),
    Price DECIMAL(10,2),
    StockQuantity INT
);

DESCRIBE Products;
```

Expected output: DESCRIBE Products;

Field	Type	Null	Key	Default	Extra
ProductID	int	NO	PRI	NULL	auto_increment
ProductName	varchar(50)	NO		NULL	
Category	varchar(30)	YES		NULL	
Price	decimal(10,2)	YES		NULL	
StockQuantity	int	YES		NULL	

Explanation: ProductID is the primary key and is generated automatically. ProductName cannot be NULL. Category, Price, and StockQuantity are allowed to contain NULL values because no NOT NULL constraint was specified for them.

Task 2: Data Insertion (INSERT)

Q2. Insert the four products and verify the records.

```
-- Insert one product
INSERT INTO Products
    (ProductName, Category, Price, StockQuantity)
VALUES
    ('Laptop', 'Electronics', 55000.00, 15);

-- Insert the remaining three products in one statement
INSERT INTO Products
    (ProductName, Category, Price, StockQuantity)
VALUES
    ('Wireless Mouse', 'Electronics', 800.00, 50),
    ('Office Chair', 'Furniture', 4500.00, 20),
    ('USB Cable', 'Electronics', 250.00, 100);

-- Verify all inserted records
SELECT * FROM Products;
```

Expected output: SELECT * FROM Products;

ProductID	ProductName	Category	Price	StockQuantity
1	Laptop	Electronics	55000.00	15
2	Wireless Mouse	Electronics	800.00	50
3	Office Chair	Furniture	4500.00	20
4	USB Cable	Electronics	250.00	100

Expected ProductID values assume a fresh table and inserts executed in the order shown. AUTO_INCREMENT values may differ if the table was previously used.

Task 3: Data Modification (UPDATE)

Q3. Update the Wireless Mouse price and increase stock for Electronics products.

```
-- 1. Set Wireless Mouse price to 950.00 using its ProductID
UPDATE Products
SET Price = 950.00
WHERE ProductID = 2;

-- 2. Add 10 units to every Electronics product
UPDATE Products
SET StockQuantity = StockQuantity + 10
WHERE Category = 'Electronics';

-- 3. Verify the updated records
SELECT * FROM Products;
```

Expected output after updates:

ProductID	ProductName	Category	Price	StockQuantity
1	Laptop	Electronics	55000.00	25
2	Wireless Mouse	Electronics	950.00	60
3	Office Chair	Furniture	4500.00	20
4	USB Cable	Electronics	250.00	110

Important: ProductID = 2 is correct only if the table is fresh and the records were inserted in the stated order. If your IDs differ, first run SELECT * FROM Products and use the actual ProductID for Wireless Mouse.

Task 4: Data Deletion (DELETE)

Q4. Delete Office Chair, then delete all products priced below 300.00.

```
-- 1. Delete Office Chair
DELETE FROM Products
WHERE ProductName = 'Office Chair';

-- 2. Delete products with a price below 300.00
DELETE FROM Products
WHERE Price < 300.00;

-- 3. Verify the final remaining records
SELECT * FROM Products;
```

Expected final output:

ProductID	ProductName	Category	Price	StockQuantity
1	Laptop	Electronics	55000.00	25
2	Wireless Mouse	Electronics	950.00	60

Office Chair is removed by the first DELETE statement. USB Cable is removed by the second statement because its price is 250.00, which is less than 300.00.

Task 5: Conceptual & Error Analysis

Q1. What happens if you execute an UPDATE statement without including a WHERE clause?

Answer: If an UPDATE statement is executed without a WHERE clause, all rows in the table are updated. A WHERE clause should be used when only specific records need to be changed.

```
-- Example (do not run unless you intend to update every row)
UPDATE Products
SET Price = 1000.00;
```

Q2. What error code or message do you receive if you attempt to insert a duplicate value into a PRIMARY KEY column?

Answer: MySQL returns error code 1062 (SQLSTATE 23000), indicating a duplicate entry for a primary key. The exact wording and key name may vary by MySQL version.

```
-- Example: this fails if ProductID 1 already exists
INSERT INTO Products
    (ProductID, ProductName, Category, Price, StockQuantity)
VALUES
    (1, 'Keyboard', 'Electronics', 1200.00, 30);

ERROR 1062 (23000): Duplicate entry '1' for key 'products.PRIMARY'
```

Conclusion

In this lab, a Products table was created and its structure described. Four products were inserted, selected for verification, updated using conditional statements, and deleted according to the given conditions. The exercise demonstrates basic DDL and DML operations in MySQL, along with the importance of WHERE clauses and primary-key uniqueness.

Submission reminder: The output tables in this PDF are expected results, not screenshots of a live MySQL session. If your instructor requires actual execution evidence, run the SQL on your system and attach screenshots of the outputs.